CASE STUDY 5

Password Policy Validation

Python Activity-Based Learning (ABL)

1. Problem Statement

Write a Python program to check whether a password follows a simple password policy. A valid password must contain at least 8 characters, at least one uppercase letter, at least one lowercase letter, and at least one digit.

2. Objectives

• To learn string handling in Python.

• To practise if/else conditions.

• To check password length and character types.

• To understand basic password rules.

3. Assumptions

• The user enters a password as text.

• The password must have at least 8 characters.

• It must contain one uppercase letter, one lowercase letter, and one digit.

• This is a classroom example, not a complete real-world security checker.

4. Algorithm

1. Start the program.

2. Read a password from the user.

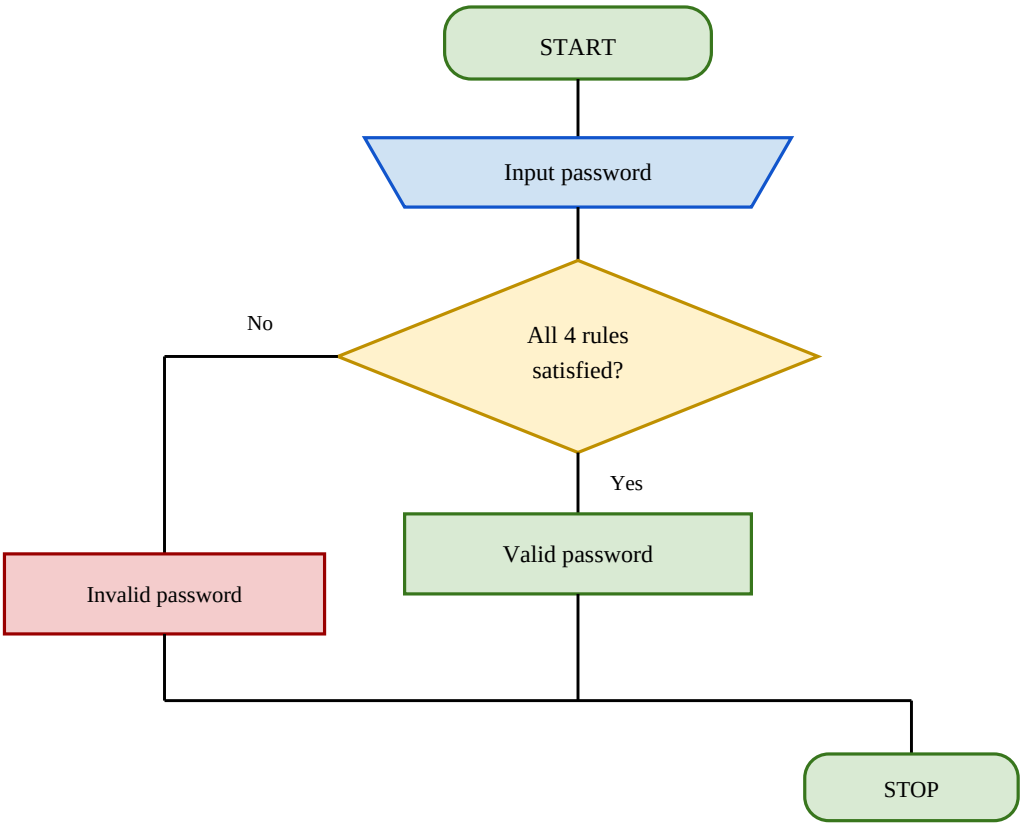
3. Check whether its length is at least 8.

4. Check whether it has an uppercase letter, a lowercase letter, and a digit.

5. If all rules are satisfied, print “Valid password”. Otherwise, print “Invalid password”.

6. Stop.

5. Colour Flowchart



6. Pseudocode

```
START
INPUT password
SET has_upper = False
SET has_lower = False
SET has_digit = False

FOR each character in password
    IF character is uppercase: has_upper = True
    IF character is lowercase: has_lower = True
    IF character is a digit: has_digit = True
END FOR

IF length(password) >= 8 AND has_upper AND has_lower AND has_digit
    PRINT "Valid password"
ELSE
    PRINT "Invalid password"
END IF
STOP
```

7. Python Source Code

```
password = input("Enter password: ")

upper = False
lower = False
digit = False

for ch in password:
    if ch.isupper():
        upper = True
    if ch.islower():
        lower = True
    if ch.isdigit():
        digit = True

if len(password) >= 8 and upper and lower and digit:
    print("Valid password")
else:
    print("Invalid password")
```

8. Test Cases

Test	Input password	Expected output
1	Hello123	Valid password
2	hello123	Invalid password (no uppercase letter)
3	HELLO123	Invalid password (no lowercase letter)
4	HelloWorld	Invalid password (no digit)
5	Ab1	Invalid password (less than 8 characters)

9. Sample Output (Output Screenshot Reference)

```
Enter password: Hello123
Valid password
```

This is the expected output for the sample input. For your final submission, run the code in VS Code and take a screenshot showing your own code and output.

10. Conclusion

This program checks a password using length, uppercase, lowercase, and digit rules. It helped me practise loops, string methods, Boolean variables, and if/else conditions.